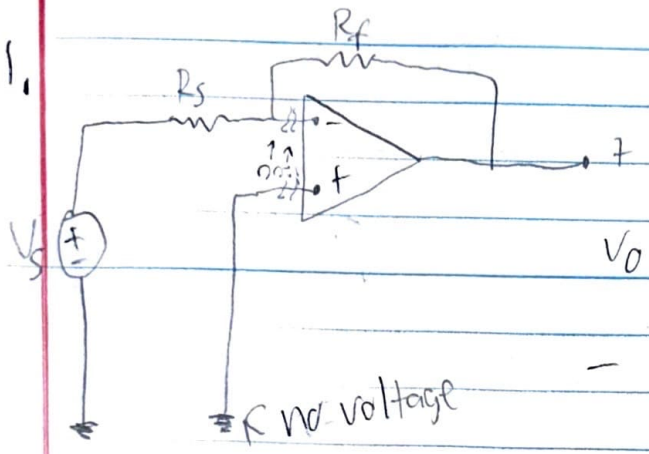
Prelab 8

Dylan Thornton

q1.



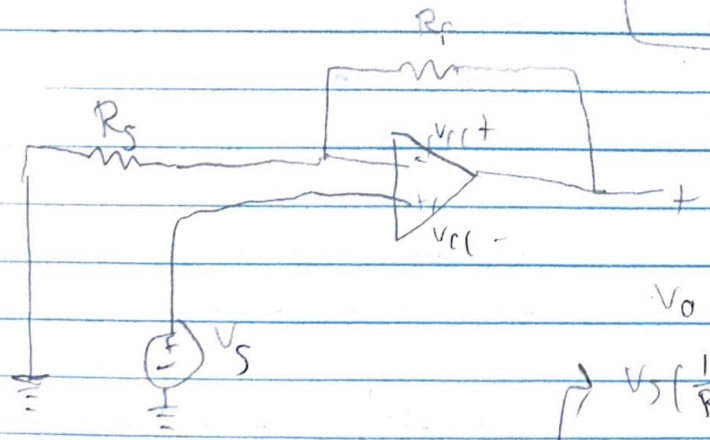
$$\frac{0 - V_s}{R_s} + \frac{0 - V_o}{R_f} = 0$$

$$\frac{-V_s}{R_s} - \frac{V_o}{R_f} = 0$$

$$\frac{-V_s}{R_s} = \frac{V_o}{R_f} \rightarrow \boxed{-\left(\frac{R_f}{R_s}\right) V_s = V_o}$$

$$\boxed{-\frac{R_f}{R_s} = \frac{V_o}{V_s} = \text{Gain}}$$

q2.



$$\frac{V_s}{R_s} + \frac{V_s - V_o}{R_f} = 0$$

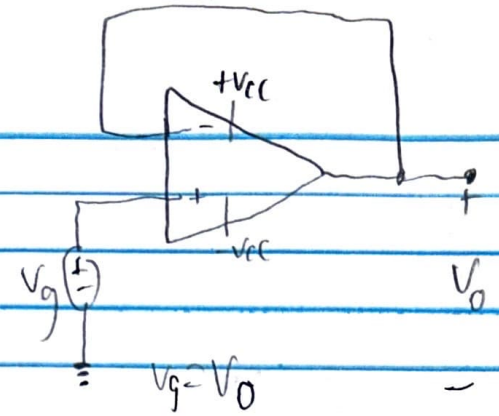
$$\frac{V_s}{R_s} + \frac{V_s - V_o}{R_f} = 0 \rightarrow V_s\left(\frac{1}{R_s} + \frac{1}{R_f}\right) - \frac{V_o}{R_f} = 0$$

$$V_s\left(\frac{1}{R_s} + \frac{1}{R_f}\right) = \frac{V_o}{R_f}$$

$$\boxed{V_s\left(\frac{R_f}{R_s} + 1\right) = V_o}$$

$$\boxed{\frac{R_f}{R_s} + 1 = \frac{V_o}{V_s} = \text{Gain}}$$

a3.



$V_g = V_o$

$V_g - V_o = 0$

$$\boxed{\begin{gathered} V_o = V_g \\ \frac{V_o}{V_g} = 1 = \text{gain} \end{gathered}}$$

b.

$-15 \le V_o \le 15$

$$\frac{0 - V_s}{R_s} + \frac{0 - V_o}{R_f} = 0$$

$\text{Gain} = -\left(\frac{R_f}{R_s}\right)$ $\quad R_f = 10k$, $R_s = 1k$ $\quad -\left(\frac{R_f}{R_s}\right) = -\left(\frac{10k}{1k}\right) = -10$

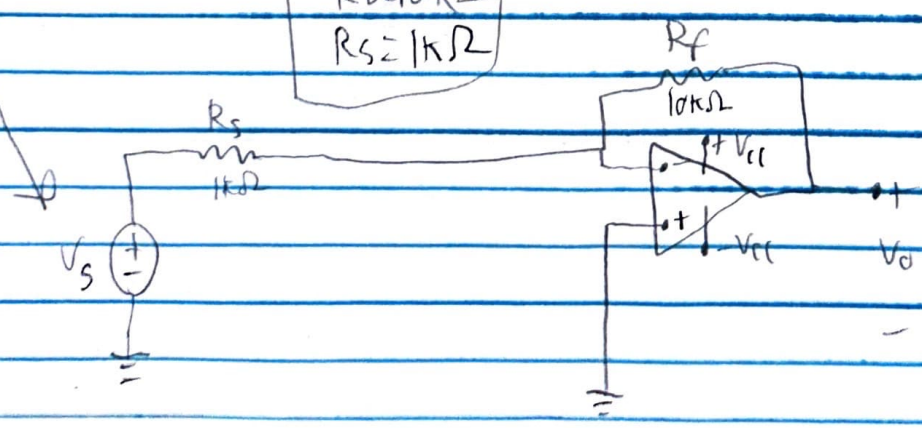
gain = 10

$15 = 10(\text{max voltage})$

$1.5 = \text{max input}$

$$\boxed{\begin{gathered} 1.5\,V \\ R_f = 10k\Omega \\ R_s = 1k\Omega \end{gathered}}$$

$-1.5 \le V_s \le 1.5$

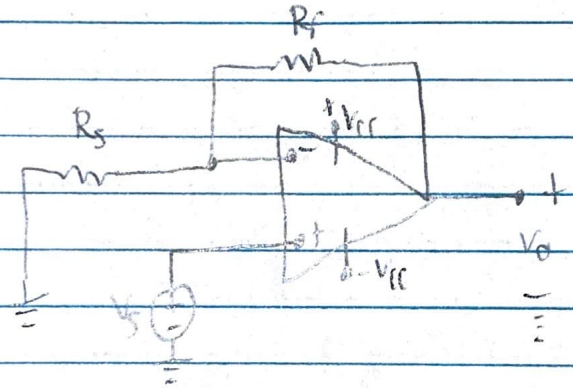


c. Gain = 5

$5 = \frac{R_f}{R_s} + 1$

$4 = \frac{8k}{2k}$

$R_f = 8k\Omega$ $V_s = 3v$ for max magnitude
$R_s = 2k\Omega$ in linear region

$15 = 5$ (max input)
$3 =$ max input

$-3 \leq V_s \leq 3$